

Jacob Sanders

Applied AI Engineer

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SUMMARY

Applied AI engineer and two-time founder shipping production systems end-to-end, from board bring-up to cloud-native SaaS, over 13+ years. Thrives in ambiguous, high-ownership environments: scopes vague problems, ships working software in weeks, and iterates directly with users. Rare breadth-plus-depth profile, full-stack (React, Next.js, TypeScript, serverless AWS) built on deep systems foundations (bare-metal C/C++, BLE, blockchain infrastructure), with fast domain ramp across wildly different verticals. AI-native builder shipping production LLM applications on the Claude API: multi-agent pipelines, structured extraction, semantic search, guardrails, and leak-free evaluation harnesses.

EXPERIENCE

Lien Into Revenue Services, LLC

Co-Founder

Huntington Beach, CA (Remote)

August 2025 – Present

- Co-founded and developed a production SaaS platform that manages the full lifecycle of personal injury chiropractic cases, from patient intake through lien negotiation and settlement collection
- Built the frontend in Next.js (App Router) and React, delivering a responsive role-based UI for chiropractors and case managers
- Designed the serverless AWS backend integrating DynamoDB, Lambda, AppSync (GraphQL), Cognito, S3, and CloudFront via AWS Amplify
- Implemented lien negotiation tooling enabling users to track outstanding liens, manage settlement offers, and monitor collection status across active cases

ARK Electronics

Full-Stack Engineer (1099 Contract)

Remote

February 2025 – April 2025

- Embedded with the client to scope requirements from scratch and ship a cloud-based manufacturing traceability platform end-to-end in weeks, covering product setup, serial number generation, barcode artifact creation, and test-result tracking for production and QA workflows
- Built a React/TypeScript/Vite (CoreUI) frontend and a fully serverless AWS backend using Lambda, AppSync (GraphQL), API Gateway (REST), DynamoDB, S3, and Cognito
- Exposed dual API surfaces (GraphQL for the web UI, REST for external clients) with API key management, usage plans, throttling, and event-driven job status updates for long-running generation batches

The Honeydew App, Inc.

Founder

Salt Lake City, UT (Remote)

April 2023 – February 2025

- Founded an on-demand handyman services marketplace; owned all system architecture and the majority of engineering while managing 4 University of Utah CS interns across frontend and backend workstreams
- Built the consumer-facing Flutter mobile application using an MVVM architecture backed by a serverless TypeScript/Supabase backend covering authentication, real-time job dispatching, and data persistence
- Led UI/UX design in Figma in collaboration with an intern designer, producing complete user flows and high-fidelity mockups; delivered ~90% MVP completion before redirecting focus to a new venture

Nerd United

Blockchain Engineer

Lehi, UT (Hybrid)

April 2022 – April 2023

- Developed blockchain applications in Go and Solidity on EVM-based protocols for cryptocurrency clients; deployed 8 L1 GETH-based Proof-of-Authority testnets and 4 production mainnets (*technology available on request*)
- Built and deployed a production generative-AI pipeline in Python, orchestrating the OpenAI image-generation API, IPFS pinning, ERC-721 contract deployment to Sepolia Testnet, and end-to-end NFT minting as an automated multi-step workflow
- Validated smart contract behavior using the Brownie framework; implemented security controls and access enforcement across blockchain infrastructure

Skullcandy

Firmware Architect

Park City, UT

October 2018 – March 2022

- Architected firmware platform strategy across a portfolio of consumer audio products, establishing design patterns and standards that improved code quality and reduced per-product bring-up time

- Directed 3–5 firmware engineers through the full product lifecycle from prototype through design validation, production validation, and mass-production release; owned Apple MFi certification compliance and served as primary liaison to Apple’s MFi program office
- Authored a C# desktop application for USB firmware updates and a Python CLI build-environment toolchain with automated regression tests, eliminating manual flashing and compressing nightly validation cycles

Code Corporation

Senior Firmware Engineer

Draper, UT

August 2013 – October 2018

- Led 3–4 firmware engineers building advanced barcode reading systems from board bring-up through production; drove Agile/Scrum adoption within the team
- Reduced device firmware upgrade time from ~3 min to ~60 sec via USB Mass Storage optimization, a 3× improvement in field servicing throughput
- Implemented a secure bootloader and BLE firmware for the Nordic nRF52832 SoC; contributed Linux kernel patches for image sensor, GPIO/PWM, and USB Mass Storage support; implemented Apple MFi chip authentication and the iAP2 protocol stack

TECHNICAL SKILLS

Languages	Python, TypeScript, Go, C, C++, C#, Objective-C, Swift, Kotlin
AI / LLM	Claude API (Anthropic SDK), multi-agent pipelines, LLM-output evals / backtesting, structured outputs, semantic search (embeddings), guardrails, model tiering & cost optimization
Web / Mobile	Next.js, React, Flutter (MVVM), Vite, GraphQL, Figma
Cloud / Backend	AWS Amplify, Lambda, DynamoDB, AppSync, API Gateway, Cognito, S3, CloudFront, Supabase
Embedded / HW	BLE/Bluetooth, Apple MFi/iAP2, RTOS, SPI/I2C/UART, USB (UDC, Mass Storage, HID), bare-metal ARM
Blockchain	EVM, Solidity, Go-Ethereum (GETH), IPFS, ERC-721, Brownie
Dev Tools	Git, Docker, Bash/Shell, REST/GraphQL APIs, Atlassian Suite

PROJECTS

Sentics: Multi-Agent Crypto Signal Pipeline

Python, Claude API, Next.js, Supabase, Vercel

Serverless multi-agent pipeline screening top crypto assets: a deterministic technical-analysis stage sets directional calls, a Claude synthesis stage writes rationale under a strict output whitelist (can’t override ground truth), and a second Claude stage distills news into structured sentiment reads. Model-tiered for cost (88% run-cost reduction) with a leak-free forward-tracking eval harness scoring live calls against realized returns (edge, hit-rate, Sharpe, t-stat).

github.com/jsanders5/sentics

Resumi: LLM Resume/Job-Matching Platform

Next.js, TypeScript, Claude API, Voyage embeddings

Full-stack app that scores job postings against a resume via the Claude API (structured JSON breakdowns across skills, experience, domain, requirements) and ranks matches with semantic search over Voyage embeddings. Prompt caching, backoff retries, and Sentry observability; deployed serverless on Vercel.

github.com/jsanders5/resumi

PySando: Interactive Python Learning Platform

FastAPI, React, Claude API, pytest

Full-stack tutorial that sandboxes and grades learner code against hidden pytest suites, paired with a Claude-powered tutor grounded in the live lesson, exercise, and learner’s code, guardrailed to never reveal solutions. Async multi-turn conversation state, provider abstraction, graceful degradation.

github.com/jsanders5/PySando

EDUCATION

University of Utah

Salt Lake City, UT

M.S., Electrical & Computer Engineering

2013

B.S., Computer Engineering

2013